

Salwi Kumari

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EDUCATION

Rungta College of Engineering and Technology
Bachelor of Technology in Computer Science (Artificial Intelligence)

Bhilai, Chhattisgarh
April. 2023 – April 2027

Bradford International Business School- 81 percentage
Intermediate - 81 percentage

Patna, Bihar
April 2019 – April 2020

Vidya Sanskar School- 87 percentage
Matriculation - 87 percentage

Patna, Bihar
April 2020 – April 2022

OBJECTIVE

- Enthusiastic B-Tech (AI) sophomore with a strong foundation in front-end development (React.js, Tailwind CSS, JavaScript) and DSA in C++. Currently exploring backend technologies to become a full-stack developer. Passionate about building scalable and userfriendly web applications. Seeking an internship to apply my skills in a real-world environment and contribute effectively to a team.

Work Experience:

Remote — [June, 2025] – [June, 2025]- Developed a Python-based data analysis pipeline to reconstruct the Hubble parameter—a key metric for measuring cosmic expansion—using observational datasets (Type Ia supernovae, baryon acoustic oscillations). Implemented cosmological models to analyze redshift-distance relationships, leveraging statistical methods (MCMC sampling, curve-fitting) for parameter optimization and uncertainty quantification. Engineered data preprocessing workflows to clean, calibrate, and interpolate sparse astronomical data using Gaussian Process regression. Visualized results through Hubble diagrams and parameter contour plots, comparing empirical reconstructions against theoretical cosmological models. Collaborated remotely with PhD advisors via weekly progress reviews, addressing challenges in data noise and model convergence. Technical Toolkit: Python (NumPy, Pandas, SciPy, scikit-learn), Jupyter, Git, Linux..

Skills and Abilities:

Proficient in full-stack web development with a solid grasp of frontend technologies including HTML, CSS (Tailwind), JavaScript, and React.js, along with backend tools like MongoDB, Mongoose, Redis, and Judge0 API integration. Skilled in developing responsive, scalable, and user-friendly web applications. Strong foundation in Data Structures and Algorithms (DSA) using C++, enabling efficient problem-solving and code optimization.

PROJECTS

Blockchain based decentralized voting system | *Express.js, Node.js, MongoDB, Ethereum, smart contracts, Machine learning*

- Blockchain-based solution leveraging Solidity smart contracts on Ethereum to enhance [e.g., supply chain transparency/secure voting]. It addresses [specific issue, e.g., data tampering] by enabling immutable, decentralized record-keeping. Built with Web3.js and React, the project demonstrates my expertise in blockchain development and scalable DApp design.
- Visualized GitHub data to show collaboration

Leetcode-clone | *Express.js, Node.js, MongoDB, Redis, Judge0 API*

April 2025 – Present

- Developed a full-stack coding platform that allows users to practice coding problems and view real-time code execution using Judge0. Implemented user authentication, problem management, and code submission flow with backend caching using Redis for enhanced performance.
- Visualized GitHub data to show collaboration

DSA Visualizer | *Express.js, Ethereum*

January 2025 - present

- Worked on an AI automation blockchain-based project under Google Solutions Challenge. Designed and developed automation workflows improving efficiency by 40%. Collaborated with a team to integrate blockchain technology for secure transactions.

Chat-gpt clone | *Express.js, Node.js, MongoDB, Gemini API*

April 2025 – Present

- Created a conversational AI interface mimicking ChatGPT functionality by integrating a large language model API. Designed for real-time QA, context handling, and seamless user experience.
- Visualized GitHub data to show collaboration

Swiggy-Frontend-clone | *React.js, Node.js,Swiggy AIP*

January 2025

- Built a fully responsive Swiggy website clone using React.js Tailwind CSS Implemented an optimized UI with reusable components Improved page load speed by 30management
- Visualized GitHub data to show collaboration

TECHNICAL SKILLS

Language: Java, Python, C/C++, SQL, NoSQL,Redis, JavaScript, HTML/CSS,

Frameworks: React, Node.js, Express.js, Judge0

Developer Tools: Git, Git-Hub, Docker, VS Code

Libraries: pandas, NumPy, Matplotlib

COMMUNICATION SKILLS

- Excellent communication skills enable me to collaborate with clients, stakeholders, and cross-functional teams to deliver highquality results. In English and Hindi.
- Worked in Group projects, Organized Events in college.